

Review of Simulation Modeling Approaches in Water Resources Management

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Abstract

Over the past few decades, numerous computer simulation models have been developed to simulate various problems in water resources engineering, offering an alternative to costly and time-consuming physical models. This paper reviews research efforts related to simulation models and their applications, focusing on scientific research where these models have been implemented. The review analyzes the most important hydrological and hydraulic simulation software used in scientific research over recent decades. The research focuses on the characteristics of these models and the differences between them in terms of methodology and application. The paper provides valuable insights and guidance for researchers wishing to apply these models to better select the most appropriate model for the problems they need to solve and to enhance the accuracy of their results.

Keywords: Hydrological models; surface runoff; flood, simulation, Iber.

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1. Introduction

Modeling is a set of procedures that involve interconnected complex operations to construct a model representing a real problem by representing that real problem with a set of mathematical equations. Models can be classified into two categories. Models are divided into two categories: physical models, which are economically costly and require a long time to implement, and computer models, which reduce implementation time and cost. The objective of modeling is to analyze the behavior of the system to identify its characteristics (if the system already exists) or to determine the optimal design for it in the future [1].

The second category is mathematical modeling, which expresses the relationships among the physical variables of a system using mathematical relations, or is the formulation of a given problem in terms of mathematical relations referred to as the mathematical model. To construct mathematical models for any given problem, several steps must be

followed, including studying the problem and identifying its components, then specifying the inputs and outputs in light of the available capabilities, and determining the constraints on the problem, along with clarifying the relationships of influence among the unknowns. After all of the above are defined, the problem can be formulated in mathematical relations whose totality is termed the “mathematical model.” This model represents the problem in a mathematical form that can be solved using one of the available analytical methods, or by representing the system through a mathematical model and then calibrating the model with a sample of observed data, which helps in predicting situations outside the limits of the calibration sample, thereby reducing the economic cost required for fieldwork [2].

There are many types of mathematical models, such as linear models, which express real relationships using assumed mathematical relations built on studying and analyzing reality, and stochastic models, which are based on statistical equations. The conceptual model is considered one of the most

important types of mathematical models used, particularly in water resources engineering. A conceptual model is a simplified representation of the site to be modeled, derived from the morphometric, hydrological, hydraulic, and environmental characteristics of the study area. Many assumptions and simplifications must be introduced to obtain an adopted conceptual model capable of being efficiently used to achieve the intended purpose [3]. When a study area is heterogeneous due to a specific physical property, it can be subdivided into regions, each representing a similar physical property. These properties are expressed using simple geometric elements such as points, nodes, curves, and polygons to represent all the information that defines the conceptual model, including boundary conditions [4].

As a result of the major advances in computer technology and the availability of high-speed computers, scientists have been able to develop highly efficient computational models for simulating hydrological and hydraulic problems due to their great capacity for handling the large volume of data representing the different characteristics of a problem. Moreover, the great development in remote sensing and digital mapping provided by Geographic Information Systems (GIS) has supported the development of integrated systems for simulating various water resources problems [5].

2. Surface Runoff Models

Simulating surface runoff is one of the complex topics in hydrology because of the large number of factors that influence it. Surface runoff is affected by rainfall characteristics such as rainfall depth, duration, and spatial distribution, as well as by infiltration properties related to soil type, land use, and catchment characteristics [6].

A surface runoff model is a mathematical model that describes the relationships governing rainfall–runoff processes in catchments, and it estimates the runoff hydrograph as the catchment response to rainfall. Many surface runoff models have been developed, and most of these models are integrated with Geographic Information Systems (GIS), which play a central role in modern hydrological and hydrodynamic modelling by providing the spatial framework within which model inputs and outputs

are organized. Integrating hydrological and hydraulic models with GIS allows the creation and management of digital maps of terrain elevation (DEMs), land use, soil type, drainage networks, and other spatial layers required by the model, which improves the physical realism of surface-runoff representation and facilitates parameter estimation. GIS tools also automate many pre-processing tasks, such as delineating watersheds and sub-basins, extracting river networks, and generating spatially distributed parameter fields, thereby reducing the time and effort needed to build and update simulation models. In addition, linking model outputs to GIS enables the production of detailed maps of runoff depth, flow paths, and inundation extent, which are essential for interpreting results and communicating flood risk to decision-makers [7].

Surface runoff models are classified according to the mathematical treatment into three categories: lumped models, semi-distributed models, and distributed models. Lumped hydrological models treat the entire basin as a single unit and are typically used for rapid and effective flow forecasting. Distributed hydrological models use fine-resolution grid cells (such as two-dimensional models) to simulate the spatial variation of physical properties. Semi-distributed models simplify watersheds into smaller, more homogeneous sub-basins, offering greater ease of configuration and faster computation for large-scale modeling, but with lower resolution than distributed models. Due to the significant development in artificial intelligence algorithms at the beginning of the twenty-first century, scientific research using these algorithms in the field of water resources has increased considerably [8].

2.1 HEC-HMS model

The HEC-HMS (Hydrologic Engineering Center Hydrologic Modeling System) model is one of the most widely used hydrological simulation models and was developed by the Hydrologic Engineering Center of the U.S. Army Corps of Engineers (HEC) [9]. The model is designed to simulate surface runoff processes for a wide range of catchments, from large to small basins.

The model includes separate components to represent each part of the runoff process, such as loss

calculations, direct runoff, and baseflow. HEC-HMS adopts the lumped or semi-distributed modeling concept for constructing the mathematical model. In HEC-HMS, the catchments are represented by connecting a network of sub-basins through channel reaches. The model uses two hydrologic elements, the sub-basin and the junction, to build the basin model. It provides tools to perform short- and long-term simulations, from single-storm events occurring in one day to simulations extending over several months or even years using time series of rainfall data.

This model is very suitable for simulating large watersheds due to its ability to perform the simulation in a short time. However, one limitation is that it uses lumped or semi-distributed concepts, which implies using average values for simulation parameters such as runoff coefficients, soil and infiltration characteristics, and unit hydrograph techniques for transforming effective rainfall into runoff. Consequently, the resulting hydrograph does not fully reflect the detailed morphometric characteristics of the basin.

The first step in applying the model is to define the basin model. Basin boundaries and drainage networks are prepared using GIS tools, and the resulting files are then exported to HEC-HMS. Simulation is controlled using several parameters available in the model. HEC-HMS also includes calibration tools that help determine the optimal values of model parameters that minimize the objective function, using the Univariate Gradient search algorithm.

The model uses sensitivity testing, a scientific method that determines how a one percent change in input parameter values affects the model's output. This test is used as a preliminary step before calibrating the model to reduce the number of parameters that need adjusting, thus saving time and effort. The HEC-HMS hydrological model was used with a continuous soil moisture simulation scheme for twelve catchments in northern Spain [10]. It was found that the model can represent the hydrological behavior of basins under different climatic and physiographic conditions.

The WEPP and HEC-HMS hydrological models were applied along with GIS and remote sensing techniques to simulate surface runoff from the upper Betarani River catchment in eastern India for the period 1999-2005 [11]. In this study, spatial datasets derived from satellite images and digital elevation models were used to prepare input layers for the models.

HEC-HMS has also been applied for flood forecasting in several basins in China [12]. The watershed boundaries and drainage networks were delineated from DEM data using ArcView GIS, and basin characteristics were imported into HEC-HMS to build the simulation model. It was applied to estimate flood flows of the Johor Tenke River basin in Malaysia [13]. Calibration of the model was done using ten years of rainfall-runoff data (1997–2006), whereby initial losses, impervious areas, and lag times were adjusted during calibration to achieve optimal agreement between simulated hydrographs and observed ones.

ArcGIS 9.2 with HEC-HMS was utilized for calibration and validation of discharge data at the Ontakalio basin in Sri Lanka so as to generate a long-term discharge series [14]. The calibrated model gave runoff estimates that were consistent with observed records, hence proving its applicability in ungauged or poorly gauged basins. Two hydrological models, WBNM and HEC-HMS, were used with GIS as tools to simulate runoff within a small urban catchment found in Azaba City, located in northern Algeria. This study sought to understand how basin scale affects the shape of the runoff hydrograph while assessing how responsive the catchment is toward rainfall events [15].

HEC-HMS was also used to evaluate the potential for rainwater harvesting in semi-arid urban areas, specifically Johannesburg. The study assessed seasonal rainfall pattern influences, climate change impacts, as well as socioeconomic constraints on rainwater harvesting; it demonstrated that hydrological modeling can assist decision-making toward sustainable urban water management [15].

2.2. The SWOT model

The Soil and Water Assessment Tool (SWAT) hydrological simulation model is, in fact, one of the

most popular models applied worldwide. It was developed by the Thad Cochran Agricultural Research Service of the United States Department of Agriculture. This model consists of many different components representing various elements of surface runoff, evapotranspiration, infiltration, flow routing, and storage in ponds and reservoirs, as well as crop growth and irrigation with groundwater recharge and soil erosion.

SWAT uses lumped and semi-distributed modeling concepts to mathematically represent the basin. A drawback here is that, like HEC-HMS, it uses lumped or semi-distributed approaches, which means that spatially averaged values have to be used for simulation parameters such as runoff coefficients, soil characteristics, infiltration characteristics, and unit hydrograph-based flow routing. Therefore, the simulated hydrograph will not reflect all detailed morphometric characteristics of the basin accurately.

SWAT was used to estimate sediment transported by valleys draining into Haditha Reservoir in Iraq and their impact on the design life of that reservoir [16]. Results showed that SWAT can offer a reliable estimate of sediment yield with time variation. SWAT has been used to assess hydrological components under climate change conditions in a basin located in Ethiopia [17]. The study found that rainfall and temperature projections significantly impact runoff, evapotranspiration, and groundwater recharge. It was applied for predicting surface runoff and sediment yield in several basins, which proved that SWAT can simulate sediment production accurately [18]. Sediment yield estimated from another study conducted on a basin located within Iran confirmed the effectiveness of SWAT for quantifying sediment transport under varying land use and climatic conditions [19].

The applicability of SWAT for simulating runoff and nitrate transport in a flat agricultural Mediterranean catchment was improved by redefining the stream network within a GIS environment where data availability is limited. Results indicated that integrating this updated drainage network into SWAT improves its capability to simulate runoff as well as nitrate concentrations [20]. The effects of climate variables on surface runoff modeling were studied using a SWAT model

for the Talar Basin in Iran. It turned out that among all parameters, wind speed and solar radiation are the most sensitive, while temperature is the least sensitive to runoff estimation [21].

SWAT was used to model streamflow and sediment transport in the Ethiopian highlands, and the results showed that the model can reliably reproduce hydrological processes in mountainous areas [22]. In Iran, SWAT has been applied for ungauged basin hydrology simulation with good performance regarding monthly discharge reproduction and land-use change impacts [23]. The effect of basin scale on SWAT calibration for runoff in tropical catchments has also been studied, where it was revealed that model performance depends upon the spatial resolution of basin subdivision [24]. The applicability of SWAT for discharge and sediment load simulation in semi-arid Mediterranean region karstic basins was assessed; though the system is highly complex, the model results were rated as satisfactory [25].

Overall, the reviewed applications indicate that SWAT can reproduce streamflow and sediment yield with acceptable accuracy under a wide range of climatic and physiographic conditions, including basins in Ethiopia, Iran, the Himalayan region, and the Anjana Kadi catchment in India [17–19, 22, 23, 25, 26].

2.3. GSSHA model

The theoretical basis of the GSSHA (Gridded Surface Subsurface Hydrologic Analysis) model for simulating surface runoff and soil erosion is founded on a two-dimensional finite-difference scheme in which the study area is divided into a grid of small square cells resembling a mesh [27]. The grid is supplied with all necessary information using a set of layers created within a GIS environment and then converted into index maps that describe the physical characteristics of the study area.

Within the model, all components of the hydrological cycle are simulated, including surface runoff, infiltration, evapotranspiration, snowmelt, soil erosion, and groundwater recharge. One of the main advantages of GSSHA is that it is a fully distributed, two-dimensional hydrological model, which provides very high accuracy in simulating all

hydrological processes over the catchment. However, the model is a complex flow and requires high-performance computer specifications to complete simulations. The simulation time in the GSSHA model depends primarily on the network size. The smaller the cell size, the greater the number of cells, resulting in more accurate results, but also increasing the simulation time. The amount of time increment also affects the simulation time; a small increment is necessary to ensure the stability of the required calculations, but it also increases the simulation time. The duration of the rainstorm also impacts the simulation time.

Another limitation is the relatively small number of users due to the high cost of the model, the demanding computational requirements, and the complexity of model setup, which has led to a limited number of published research applications [28]. A comparison between SWAT and GSSHA for predicting surface runoff and sediment concentrations in small catchments showed a close agreement between the results of both models for long-term simulations [29].

GSSHA has been tested on a set of basins in the United States under different flow conditions, and the findings indicated that the model exhibits good efficiency in simulating runoff for these basins [29]. In another study, GSSHA, as a fully distributed model, was compared with HEC-HMS, as a lumped model, to analyze the impacts of changing the land-use on surface runoff. The results showed that GSSHA has superior capability in identifying the modification of land-use change on runoff compared with HEC-HMS [30].

3. Flood Wave Routing Modelling

The applications of hydrological simulation models are fundamental tools for planning and designing basic infrastructure in developed cities. They support measures to confront and reduce the impacts of extreme meteorological events, such as severe rainstorms. These models are used both in designing of stormwater drainage networks and in analyzing the flood disasters caused by dam failures. Flood-inundation maps produced after modelling and simulating the flood wave are among the key outputs that enable decision-makers to take appropriate

actions to limit or minimize the damage caused by such floods. Therefore, the accuracy of simulation results is an essential and critical requirement for obtaining reliable analytical outcomes.

Several computer models have been developed to simulate flood waves and to extract all details of the inundation areas that will result from flooding. Some of these models analyse the flood wave in one dimension (one-dimensional models), whereas others analyse the flood wave in two dimensions. The choice of the dimensionality of the model used depends on the morphological characteristics of the study area. When the flow is confined by mountains or any barrier that forces the flood to move in a direction normal to the cross-sectional area of the channel and prevents it from spreading in all directions, a one-dimensional model is preferred. In contrast, when the floodplain is wide and floodwaters are expected to spread in all directions, it is preferable to use a computer model that analyses the flood wave in two dimensions [31]. The most prominent computer models used to simulate flood waves include the following.

3.1. DAMBRK model

The DAMBRK model is a one-dimensional hydrodynamic computer model whose name is an abbreviation of “Dam Break Forecasting Model”, used for routing flood waves. The DAMBRK model simulates flood waves resulting from severe rainstorms or from failures that occur in dams and bridges, and it evaluates the effect of reservoir storage volume on the areas that will be inundated.

The model was developed by the U.S. National Weather Service for the purpose of forecasting flood hydrographs and routing the flood wave along the flood channel. DAMBRK is based on solving the one-dimensional Saint Venant equations for unsteady flow to determine flood characteristics, water depths, velocities, and discharges. The model uses the finite difference method as a numerical technique to solve these equations [32].

The DAMBRK model is capable of simulating the following cases:

- Simulating the effect of bridges on flow in rivers.
- Simulating the effect of meandering rivers on wide floodplains.

- Simulating flow under different hydraulic regimes, including supercritical flow, subcritical flow, and transitions between them along the river reach.
- Routing any inflow hydrograph through the downstream river system and determining the resulting water levels and inundation extent over the floodplain.
- Routing controlled reservoir outflows (without dam failure), as required in dam design and operation studies, to assess downstream water levels and inundation under planned release scenarios.
- Simulating the effects of backwater generated by dam failure at the confluence of a tributary or canal with the main river.

One of the main limitations of this model is that it is one-dimensional, which means it can only simulate variations of flow along the main channel axis and cannot represent the full lateral spreading of floodwaters on very wide floodplains. Therefore, it is less suitable for analysing flood waves in large valleys with extensive inundation areas, where a two-dimensional model that resolves flow in both longitudinal and transverse directions is preferable [33].

The DAMBRK model was used to simulate the hypothetical failure of Mosul Dam, northwest the city of Mosul in northern Iraq, to derive flood inundation maps for the resulting flow for several storage elevations [34]. A flood risk analysis was carried out by simulating floods generated by discharges released from Paldang Dam, located upstream of Seoul in South Korea [35].

3.2. HEC-RAS model

The HEC-RAS model is one of the most important and widely used hydrodynamic models for simulating water movement in rivers and channels. It was developed by the Hydrologic Engineering Center (HEC) of the U.S. Army Corps of Engineers, and the term RAS is an abbreviation of “Rivers Analysis System”. The first version of this model, HEC-RAS 1.0, was released in 1995 [36].

HEC-RAS is an integrated software system designed to operate in a multitasking computing environment and includes a graphical user interface

(GUI) that provides commands for hydraulic analysis of flow in channels, data storage and management, and graphical display of results. It simulates and analyses water movement for steady flow conditions as well as unsteady flow, and it can model sediment transport in rivers and channels in addition to water quality processes.

The model analyzes various flow conditions after these have been defined as inputs, together with all engineering details of the case under study, including discharges, cross-sectional geometry, roughness coefficients, and bed slope. Calculations are performed using standard hydraulic equations for open-channel flow, based on the principles of energy and momentum conservation, and the Saint Venant equations are solved numerically using the finite difference method (FDM) [37].

All HEC-RAS versions from HEC-RAS 1.0 (1995) to HEC-RAS 4.1 (2010) performed one-dimensional flow analysis only. The ability to perform two-dimensional unsteady-flow simulations was added in HEC-RAS 5.0 (2016), enabling the model to analyse flow in wide channels and floodplains, especially when simulating major floods where water spreads in all directions [38].

HEC-RAS is distinguished by its ability to simulate the following cases:

- Flow through single and multiple bridges and culverts.
- Flow through spillways and weirs.
- Floodplain encroachment and floodplain inundation.
- Local scour around bridge piers.
- Sediment transport in channels and inundated areas.

HEC-RAS has a clear and user-friendly graphical user interface that allows users to define geometry, flow data, and boundary conditions through interactive windows, tables, and plots. The software provides integrated tools for importing channel cross-sections from GIS, visualizing the river network, and displaying water surface profiles and inundation results, which simplifies the data-entry and model-setup process. In contrast, the DAMBRK model relies mainly on text-based input files with limited graphical support, so preparing and checking

the required data are generally more time-consuming and error-prone. Several applications in the literature emphasize that the HEC-RAS interface is easier to use and more intuitive than that of DAMBRK, particularly for complex river systems and floodplain studies [39]. The hydrological model HEC-HMS, together with HEC-GeoHMS (Hydrologic Engineering Center – Geospatial Hydrologic Modeling Extension), was used to forecast floods in the Misai and Wan'an basins in southern China, with areas of about 797 km² and 869 km², respectively [40].

Surface runoff generated by rainfall and the corresponding flood-inundation maps were simulated for a 15.92 km² study area in Illinois, USA, using HEC-HMS and HEC-RAS [41].

The two-dimensional HEC-RAS model was applied to derive inundation maps for the hypothetical failure of Ujjani Dam, located upstream of the city of Pandharpur in India [42]. Flooding in the Segamat River basin in Malaysia was modelled using HEC-RAS for different scenarios based on peak discharges associated with return periods of 10, 25, 50, 100, 200, 500, and 1,000 years [43].

HEC-RAS was used to generate inundation maps for the failure of the Baeksan levee on the Nam River in North Korea, which collapsed in 2002 due to continuous heavy rainfall, flooding six downstream villages over a total area of 3.5 km² [44]. The HEC-RAS model was combined with an artificial neural network to evaluate the accuracy of integrating both models in predicting a flood risk map for the Baro Akobo River basin in southwestern Ethiopia [45]. The Osage River downstream of Bagnell Dam in Missouri, USA, was modeled using HEC-RAS to assess bank stability under variable releases from the dam's power generation units [46 - 48]

3.3. MIKE model

The MIKE model is one of the most renowned hydrodynamic models worldwide, owing to its extensive capabilities for analysing different types of flow under a wide range of hydraulic conditions. It was developed by the Danish Hydraulic Institute (DHI), which specializes in hydraulic modelling. The model employs the finite difference method as a numerical technique to solve the Saint Venant

equations used for calculating flow parameters such as velocity, depth, and water surface elevation [49]. The most prominent MIKE products are described below.

3.3.1 MIKE 11

MIKE 11 is a one-dimensional model that simulates hydrodynamics and water quality in addition to sediment transport in natural rivers and artificial channels. Although MIKE 11 is primarily a hydrodynamic model, it can also be used in rainfall-runoff studies by coupling its channel flow module with an external or built-in rainfall-runoff component, such as the NAM conceptual rainfall-runoff model of the MIKE system, to transform rainfall into catchment inflows at the upstream boundaries [50]. MIKE 11 is capable of:

- Modelling flow in branched networks and looped channel systems.
- Modelling flood flows along riverbanks.
- Simulating both steady and unsteady flow conditions.
- Simulating water movement around bridge piers and through culverts, weirs, and regulators.
- Simulating failure processes in dams.

One of the main limitations of MIKE 11 is its limited capability to accurately simulate flooding in very wide channels and basins [51].

The hypothetical failure of Hirakud Dam on the Mahanadi River in Odisha State, India, was simulated using MIKE 11, assuming the probable maximum flood as inflow to the reservoir and failure due to overtopping [52].

3.3.2. MIKE 21

MIKE 21 is a two-dimensional model used to simulate surface water flow. It provides more detailed analyses of water movement than MIKE 11 and is employed to simulate flow in wide basins that cannot be modelled with sufficient accuracy by MIKE 11.

In MIKE 21, the study area is discretized into a grid of rectangular cells, and the model computes water levels, depths, and discharges within these cells after being supplied with the essential inputs, including topography, roughness coefficients, initial and boundary conditions. Additional forcings, such as wind speed and direction, can be specified when wind

effects on water levels or currents are important for the study objectives. The model produces time series of water depth and discharge as well as complete inundation maps for the study area, such as water-level and depth maps for different times. A key limitation of MIKE 21 is its limited accuracy in simulating water movement around buildings and complex man-made structures [53].

In [54], the study simulates floods with 50, 100, and 500-year return periods for the Khalkai River watershed located in Gilan Province, northern Iran, to assess floodplain risk, flood depth, and extent in an agriculturally and urbanized watershed that floods frequently.

3.3.3. MIKE FLOOD

MIKE FLOOD is one of the tools within the MIKE modelling system for simulating water movement under various flow conditions. It couples one-dimensional (1D) and two-dimensional (2D) models into a single framework, providing great flexibility in dealing with a wide range of modelling scenarios and avoiding many of the constraints imposed by using each model separately [55]. Its main applications include:

- Simulation of ordinary flood events.
- Storm-rainfall and pluvial flood studies.
- Simulation of dam-break floods.
- Supporting the hydraulic design and evaluation of different water structures, such as spillways, culverts, levees, and flood protection works, by simulating water levels, flow velocities, and inundation patterns for alternative design scenarios.
- Modelling of various estuarine and river-mouth conditions.

MIKE FLOOD was used to obtain inundation maps and flood levels in the delta of the Mahanadi River basin in India, which covers an area of 6,800 km² [56]. The MIKE FLOOD 1D–2D hydrodynamic model was applied to derive flood maps generated by storm events with return periods of 5, 20, 50, and 100 years for the Scotchman's Creek catchment in Melbourne, Australia. The objective was to exploit floodwater for rainwater-harvesting purposes in natural depressions acting as reservoirs, thereby

reducing pollution from storm discharges and lowering flood risk [57]. MIKE FLOOD was also used to simulate the hypothetical failure of the DakDrinh Dam on the Khuc River in Vietnam [58].

In another study, MIKE FLOOD and ArcGIS were jointed to predict flood-induced damages in the densely populated coastal area of Musgrave Avenue in Australia, which covers an area of 1.82 km² [59]. The study [60] proposed a method to boost the performance of the MIKE FLOOD 1D–2D model by combining a detailed evaluation of the digital elevation model (DEM) with satellite-based validation of simulated flood extents in the lower Chao Phraya basin in Thailand. [60].

3.4. Iber model

The Iber is a modern hydrodynamic model that simulates the movement of shallow water and is a two-dimensional computer model developed by the Hydraulic Studies Centre at the University of A Coruña in Spain. The model was named after the Iberian Peninsula, located between Spain and Portugal, in recognition of the joint scientific effort of research teams from both countries in developing it.

Iber consists of an integrated suite of applications designed to simulate turbulent, unsteady surface flow and sediment transport under shallow-water conditions. The Saint Venant equations are used to evaluate flow depth and velocity in shallow flow, and Iber uses the finite volume process to solve these equations [61].

The most important types of problems that can be simulated using Iber include:

- Simulation of water movement in streets and road intersections.
- Simulation of flooding in streams and rivers due to large discharges and derivation of inundation maps.
- Simulation of the hydraulic effects of river structures such as bridges, groynes, and culverts.
- Simulation of hypothetical dam failures.
- Simulation of chemical pollutant transport in rivers.
- Simulation of floods generated by storm rainfall and estimation of outflows from catchments.

- Simulation of wind effects on water movement in reservoirs and rivers.

The hypothetical failure of Chipembe Dam in Mozambique was simulated using the IBER model to obtain flood-inundation maps extending 36 km downstream of the dam [62]. Another study investigated the capability of Iber to simulate different flood scenarios and compared its performance with other computer models. Four models, Tuf low FV, Info Works ICM, JFlow+, and Iber, were tested by simulating several flood scenarios, including filling a water body of known dimensions, filling a depression with complex topography, and simulating a flood in a known valley [63].

The Iber hydrodynamic model was also used to simulate the instantaneous hypothetical failure of Mosul Dam, located 50 km northwest of the city of Mosul in northern Iraq [64]. The study [65] assesses flood hazard in Driouch City, northeastern Morocco, by comparing a GIS-based Analytic Hierarchy Process (AHP) approach with the Iber model. In [66], authors compare two 2D models, HEC-RAS and IBER, for simulating flood behavior in Wadi Deffa across El Bayadh City in Algeria to estimate flooding and hazard maps.

3.5. FLO-2D model

FLO-2D functions as a two-dimensional (2D) flood simulation program that optimizes the representation of hydrodynamic operations in urbanized and natural environments. Finite difference methods within the model solve the complete dynamic wave momentum equation, which delivers reliable simulations of flood waves throughout varied and intricate terrain features. Through its advanced features, FLO-2D effectively models different flooding scenarios, which include river overbank flooding, urban inundation, and coastal surge events, making it a vital tool for flood risk management during the emergency response planning period. Edits show top performance because it recognizes multiple flood conditions ranging from unconfined alluvial fan movements through storm drain interfaces and urban street water accumulation. The combined processes of hydrological and hydraulic modeling in the FLO-2D model operate from a unified framework, thus

eliminating the requirement for multiple rainfall-runoff and flood routing models, which results in an efficient approach for complete flood risk evaluation [67].

Authors in [68] present a physics-based framework that couples the surface water model FLO-2D with the HEC-RAS model to simulate how surface and subsurface water interact during compound flood events in a coastal basin in North Miami. The article [69] assesses debris flow risk in three gullies of Yunlong County, northern Yunnan, using the FLO-2D numerical model under extreme rainfall, and evaluates how existing and proposed engineering measures change that risk pattern before and after control.

4. Discussion

The scientific review of the aforementioned hydrological models demonstrates their importance in conducting hydrological studies and research. These models can be classified, according to their mathematical formulation, into lumped models, semi-distributed models, and fully distributed models.

From the analysis of the results reported by different researchers, it is evident that fully distributed models provide the highest accuracy; however, they require very powerful computers to run the simulations or even to operate the model itself. Lumped models are the most widely used because they are mathematically less complex, do not require high-specification computers, and have very short simulation times, which makes them particularly suitable for studies in large catchments.

Semi-distributed models represent an intermediate case between fully distributed and lumped models; they are more accurate than lumped models but less accurate than fully distributed models. From the analysis of surface runoff simulation models, it appears that SWAT is the most substantially used hydrological model worldwide and is a lumped and semi-distributed model similar to HEC-HMS.

The review also indicates that GSSHA is the best hydrological model in terms of physical representation, as it is a fully distributed model with very high capabilities, allowing the study area to be

simulated in a way that closely reflects reality. However, its use is still very limited because of its relative complexity and the need for advanced computer hardware to run the simulations.

Analysis of flood simulation models reveals that most of them rely on the Saint-Venant equation as the mathematical basis for flood simulation. Flood simulation models differ in their scale, ranging from one-dimensional to two-dimensional. A review of previous studies indicates that the MIKE model

group is among the most widely used flood simulation models globally, compared to the HEC-RAS model and the Dam Break model. The Iber model is a modern, high-specification hydrodynamic model that uses finite volume as its mathematical basis. This type of mathematical processing yields more accurate results than finite elements and finite differences. However, its global user base and the number of research studies employing it are limited. Table 1 shows a comparison between the models that were discussed in the research.

Table 1: Comparison between the hydrological models.

Model	Developer	Mathematical Foundation	Dimensions	Users worldwide
HEC-HMS	Hydrologic Engineering Center, U.S. Army Corps of Engineers	Lumped	Single block or half distributor	Many
SWAT	USDA Agricultural Research Service	Lumped	Single block or half distributor	Many
GSSHA	U.S. Army Corps of Engineers	Finite difference	2D	Few
HEC-RAS	Hydrologic Engineering Center, U.S. Army Corps of Engineers	Lumped	1D	Many
Dam break	National Weather Service	Lumped	1D	Many
MIKE	Danish Hydraulic Institute DHI	Finite difference	1D or 2D	Many
Iber	Hydraulic Studies Centre at the University of a Coruña in Spain	Finite volume	2D	Few
FLO	FLO-2D Software, Inc., Nutrioso, Arizona, USA	Finite difference	2D	Few

5. Conclusions

Hydrological and hydraulic simulation models are essential computer tools to understand, analyze, and predict water movement. These models assist engineers and scientists in water resource management, infrastructure design, and flood risk assessment. Hydrological simulation models focus on water volume and runoff from rainfall and how it moves within watersheds. Two well-known examples of hydrological simulation models are HEC-HMS and SWAT. Hydraulic simulation models, on the other hand, focus on water movement (velocity, depth, and direction) within a watercourse or structure. Two popular programs used for this purpose are HEC-RAS and MIKE FLOOD. The selection of the appropriate model for a given

problem, whether hydrological or hydraulic, depends on the simulation objective, parameter, and problem nature.

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Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this article.

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